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PRINCIPLES OF PERFORMANCE [1]

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Apparently it behooves a so-called "cognitive theorist" such as myself to explain how his assumed intervening cognitive and motivational variables ever issue into actual behavior. I must, it seems, provide some *rules* or *principles* of *performance* according to which these variables—namely, discriminations, expectancies, drives, incentive values—add to or multiply with one another (or whatever the appropriate combining operations may be) to produce behavior. Otherwise, as Guthrie (5) has suggested, I seem to leave my animal (whether rat or human being) merely "buried in thought" (though why this should be considered so shocking in the case of a human being I have never quite been able to fathom).

Let me begin by summarizing briefly my basic assumptions. First, I assume four distinctive types of *independent variables:* (*a*) present stimulus-units; (*b*) previous presentations of stimulus-unit–response–stimulus-unit sequences, abbreviated as past (*S-R-S*)'s; (*c*) present drive disturbances and currently presented incentive objects; and (*d*) genetic or early training differences between individual subjects (as parameters affecting the results from the other three types of independent variable).

[1] Presented, with some modification, as presidential address to the Division of General Psychology of the American Psychological Association, Sept. 7, 1953.

Secondly, I assume three types of *intervening variables* and *intervening-variable systems:* (*a*) discriminations within a discrimination system; (*b*) beliefs and expectancies within a belief-expectancy system; and (*c*) drives and incentive-values within a drive-incentive-value system. The three intervening variable systems are inferred sets of dispositions (in the sense of propensities of the organism to perform in certain prescribable ways under specified test conditions). And I assume that the momentary activated discriminations, beliefs, expectations, drives, and incentive values are determined by the momentary values of the presented independent variables in the contexts of these three innate and acquired dispositional systems.[2] Let me elaborate.

1. *The Discrimination System.* This is to be specified by (*a*) *categorizing* or *nominal dimensions,* that is, types of general object-classification, of which the given organism has become capable (e.g., most human beings have acquired

[2] I wish to express my great indebtedness to Professor Leo Postman for his unstinting and crucial help in the development of these concepts concerning the independent and the intervening variables and variable systems. The performance principles which I shall propose later in this article, though in part Professor Postman's, are at present largely mine—at least insofar as the pictorial language is concerned. Postman does not find this language helpful.

such categories as colors, pitches, smells, chairs, tables, trees, other human beings, etc., etc.), and by (*b*) the *intensitive dimensions* along which the given organism has, in addition, become capable of making intensity discriminations (e.g., specific hues, heights of pitches, intensity of odors, comfortableness of chairs, sturdiness of tables, shadiness of trees, likeableness of human beings, etc.). Such categorizing and intensitive dimensions are uncovered in standard situations in which the organism is presented with selected arrays of stimulus-units and which it is forced (or tricked) into categorizing and/or discriminating intensitively. In other words, the discrimination system is to be discovered by the methods of a comprehensive and up-to-date, as well as a mathematically sophisticated, psychophysics. Knowing this discrimination system, we can thereafter state the specific categorizations and intensity discriminations which are likely to be activated when such and such stimulus-units are actually presented on any given occasion.

2. *The Belief-Expectancy System.* This is to be specified by the sum of the organism's expectancy dispositions. For example, a subject may show that upon the sounding of a given tone, S_1, he is prepared, provided he waits, R_1, in a given place, for the coming of a shock, S_2. Or again, he may show that when presented by the choice point of a maze, S_1, he is prepared for a resultant food, S_2, provided he turns, R_1, to the right. Or he may show that upon the presentation of a first syllable, S_1, on a memory drum he is prepared, provided he continues to look at the drum, R_2, for the coming of a second syllable, S_2. Any one such set in the animal, which (to distinguish it from the objective occurrences) may be symbolized as an $(\dot{S}_1\text{-}\dot{R}_1\text{-}\dot{S}_1)$, I would designate as an activated expectation. The continued possession by the organism of such a set I would call an expectancy disposition. An "expectancy disposition" is thus a small acquired disposition of greater or less certainty and permanence, which will tend to lead to an activated expectation whenever an instance of the given sign stimulus-unit is presented. Such an expectancy disposition will also tend to be activated by other stimulus-units which are near the sign stimulus on some common dimension of generalization. Thus, there may be groups or families of expectancy dispositions in the sense that an array of them related by different signs may all have attached to them similar expectancy dispositions as to the type of significate to be reached, if a certain general type of means-manipulatory response be carried out. Such a group or envelope of related expectancy dispositions I shall call a "belief disposition." An individual's expectancy system is constituted by the sum of all his expectancy and belief dispositions.[3]

3. *The Drive-Incentive-Value System.* This system is to be conceived as a collection of dispositions to approach and/or consume specific types of positive incentive objects or to escape from and/or avoid specific types of negative incentive objects. These approach (consummation) and avoidance (and/or escape) dispositions depend upon the interaction of two distinguishable independent variables: namely, (*a*) types and amounts of drive-disturbance, and (*b*) types of stimulus-units presented as incentive objects.

a. Drive-disturbances would be: hours of specific food-deprivation, hours of sex-deprivation in males, time-position of the female in the oestrus cycle, per-

[3] I would like to emphasize that my notions about expectancies have been clarified by the recent discussion by MacCorquodale and Meehl. I wish here to acknowledge my very great indebtedness to them (13).

centage decrement in bodily weight, hours of water-deprivation, amounts and prior time of saline injections, intensities of just-preceding electric shocks, hours of social deprivation, specific raisings and lowerings of bodily health, special maintenance-schedule with respect to aesthetic objects or with respect to exercise, etc., etc.

b. Incentive objects would be: specific foods, liquids, sex objects, groups of individuals of the same or different species, barriers, and distances to be overcome, enemies, friends, present electric shocks, poems, paintings, symphonies, open windows to look through in the case of monkeys (see Harlow [6]), etc.

The character and strength of an activated approach (consummation) and avoidance (escape) disposition will result from the combination of the particular drive-disturbance and the particular incentive object presented on the given occasion. The sum total of drive-disturbance and incentive object relations obtaining for a given individual make up his drive-incentive-value system.

Assuming now that the above brief discussion of discrimination dispositions, expectancy dispositions, and drive-incentive-value dispositions has succeeded in presenting the basic theoretical position, I wish to pass on to the main topic of this paper. How on particular occasions—given the independent variables of presented sign stimulus-units, previously presented (*S-R-S*)'s, present drive-disturbances and present positive or negative incentive objects—do the to-be-inferred resulting then-and-there activated discriminations, expectancies, drives, and incentive values interact to produce actual performance? What, in short, are the Rules or Principles of Performance which must be added to the system?

PERFORMANCES AND THE BEHAVIOR SPACE

I shall illustrate these principles or rules in large part by means of what I shall call behavior-space diagrams or pictures. This notion of a behavior space stems, of course, from Kurt Lewin's concept of the life space (12). It differs from the latter, however, *first*, in that the structures and processes which Lewin put inside his "psychological person" *as part* of the life space have been, in the present analysis, taken care of by the three dispositional systems of discriminations, belief-expectancies, and drives and incentive values, together with the momentary processes activated within these systems.

Secondly, my behavior space is more frankly a merely pictorial way of representing certain relationships and less a would-be true mathematics or logic.

Thirdly, I differ perhaps from Lewin in that what I am trying to illustrate or explain by the behavior space are not concrete responses but certain generalized *performances*. A "performance," as I here use the term, is not something that can be observed on a single occasion but something that can be specified only from observing two or more concrete behaviors. For example, Tolman, Ritchie, and Kalish in one of their experiments on spatial learning (25) used a single-unit T maze with two vertically opposite, alternatively usable, starting stems. And they discovered that only by observing the responses of the animals from both stems could they say whether a rat's acquired behavior—what I am here calling his "performance"—was that of "taking right-hand turns" or that of "going to the lighted side." But, whichever it was, such an "always going to the right or "always going towards the lighted side," would

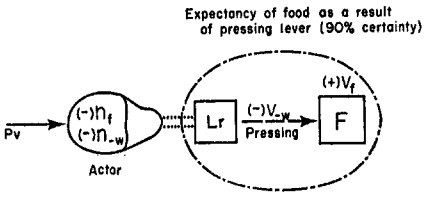


FIG. 1. Behavior-space: rat pressing lever for food in a Skinner box.

be an example of what is here meant by a performance.[4]

Fourthly, it is to be emphasized that a behavior-space diagram will be neither a picture of the actual geographical environment nor a picture of the supposed phenomenological experience of the organism. It is rather a merely pictorial way of trying to represent the interactions between the inferred activated dispositions and their consequent determination of the to-be-predicted performance. For example, my diagram for the case of the rat pressing the lever in a Skinner box (17) at a certain rate is shown in Fig. 1.

This diagram represents the assumption that the rat discriminates himself as "actor" in the presence of the discriminated lever, *Lr*. The pair of dotted lines connecting actor to *Lr* symbolizes "immediate presence of the lever." And the diagram represents further the fact that this discriminated lever, *Lr,* has become the sign term in the activated expectancy. The whole expectancy is indicated by the contents of the dash-dot ellipse. This expectancy has acquired, it is assumed, a certainty of about 90 per cent and is to the effect that if the actor responds by the "performance" of pressing, he will get to food, *F*. Also, the expected food is depicted as having a positive valence v_f, indicated by the plus sign. And the actor is shown as having an activated drive or need-push *for* food, indicated by the minus sign, labeled n_f, inside him. Further, the length of the dashed arrow, labeled "pressing," indicates the degree of effort or work which, according to the expectancy, will be involved in the pressing. And the actor is shown as having a drive or need-push *against* work indicated by the second minus sign, labeled n_{-w} within the actor. Finally, the arrow at the left, labeled *Pv,* is a vector representing the magnitude and direction of the resulting to-be-predicted performance of pressing.

Let us consider further for a moment need-pushes and valences. I conceive a need-push as, so to speak, that portion of a drive which, under the concrete stimulus conditions of the moment, gets into the behavior space. The magnitude of a need-push is, in short, assumed to be determined not only by the magnitude of the actual activated drive but also by the goodness of the discrimination of drive stimuli under the given experimental conditions.

Consider, for example, that type of latent-learning experiment which has been called by Thistlethwaite (19) the irrelevant-incentive experiment. In such experiments, during the initial training period an irrelevant goal object (say food) is presented in one alley of a T or Y maze, and water, a relevant goal object (since the animals are in this period run thirsty), is presented in the other alley. Then the rats are made hungry and it appears from the original experiment of Spence and Lippitt (18) and from various repeat experiments (see, for example, Kendler [11]) that rats often do *not* then tend to go to the food side when first made hungry. However, it has also appeared from an experiment by Thistlethwaite (20) that,

[4] For this notion of a performance as a behavioral disposition which can be identified only by observing at least two somewhat different test situations I am indebted primarily to Professor Benbow F. Ritchie. (Private communication.) But see also a quite recent statement of what seems to be essentially the same position by Campbell (2).

if the animals have had some previous training in discriminating their drive stimuli, they are much more likely to show some initial tendency to go at once to the food side. These two contrasting results would be interpreted, by me, as meaning that, although the rats have been made hungry (as would be shown by their readiness to approach or consume food in the standard situation), this hunger drive does not tend to be discriminated (i.e., to produce a corresponding need-push in the behavior space corresponding to the maze) unless there has been previous practice in drive discrimination in a mazelike situation. Further, some recent findings by Heyer and O'Kelly (7) indicate that tissue dehydration due to subcutaneous saline injection will not cause appropriate performance in a maze to get to water, *unless* this dehydration has been experienced over a considerable preceding period of time. I would interpret this also as lack of thirst need-push in the behavior space (except in the actual presence of water itself) unless there has been plenty of preceding time to experience and discriminate the special drive stimuli resulting from the saline-produced dehydration.

Return now to the Skinner box. We may probably assume that in it, because of preliminary training, the hunger drive stimuli and the nonwork drive stimuli have become completely discriminated. We may assume in short that the magnitude of the food need-push corresponds directly to the magnitude of the actual food hunger drive. And finally we may probably also assume that the magnitude of the nonwork need-push corresponds directly to the actual nonwork drive.

What now about the valences? Valences are conceived as bearing nearly the same relation to incentive-values that need-pushes bear to drives. That is, the amount of valence which the food pellets will have in the behavior-space diagram corresponding to the Skinner box will depend not only upon the incentive value of such pellets (as measured in some standard situation) but also upon the discriminability of the pellets in the concrete stimulus context of a Skinner box. And this discriminability may, hypothetically at least, be less than in the standard situation. Hence the valence of the food, in the behavior space corresponding to the Skinner box, could be less (though I doubt it) than the basic incentive value of such pellets, as measured in a standard situation. Similarly, the negative valence of the work involved in pressing might, in the behavior space corresponding to the Skinner box, be less than the actual negative incentive value of such work. But this also seems doubtful.

Look again at Fig. 1. My final argument is that as a result of (i) the need-push for food, (ii) the positive valence of the expected food, (iii) the need-push against work, and (iv) the negative valence of the expected work (this expected work is indicated by the length of the dashed arrow), there results the performance vector *Pv*. It is this vector which specifies the direction and magnitude of the to-be-predicted actual performance. The greater the valence of the expected food, the greater the food need-push, and the greater the expectancy that the food will result, the greater the magnitude of the performance vector toward actually pressing the lever. On the other hand, the greater the need-push against work and the greater the negative valence of the expected work and the stronger the expectation of this work, the less the performance vector toward pressing the lever.

For the sake, now, of the critic who dislikes such a pictorial mumbo jumbo, it may be pointed out that these relations probably can be equally well ex-

pressed by an algebraic equation (Equation 1).

$$Pv = f_x\ (n_f,\ v_f,\ \exp_f) - f_y\ (n_{-w},\ v_{-w},\ \exp_w) \quad [1]$$

Here Pv stands for the magnitude of the performance vector (i.e., the tendency to press the lever). And this magnitude is shown as increasing with the values of n_f, v_f, and $\exp_f$, combined according to some function, f_x, and as being decreased by the values of n_{-w}, v_{-w}, and $\exp_w$, combined according to some function, f_y. The exact forms of these two functions, f_x and f_y, I have not attempted to specify. It may be that these functions both have the form of simple multiplication. But in any event, it is assumed for both of these functions that the component variables all operate in a positive direction such that the increase in any one of these component variables tends to increase the value of the function.

Still further, the critic will note that the first half of the right-hand side of this equation is rather similar to Hull's equation for reaction potential, or ${}_sE_R$ (9), (which is presented here as Equation 2).

$$\text{(Hull)}\ {}_sE_R = f\ (D \times V_1 \times K \times {}_sH_R) \quad [2]$$

Hull's D, or drive, is closely similar to my n_f, or hunger need-push. His V_1, or stimulus intensity dynamism, is, however, a term I do not have. For me the effect of any *intensity* or a presented stimulus-unit such as the stimulus-pattern resulting from the lever would already have been evaluated by an analysis of the discriminatory dispositional system and of the effects in it of intense or nonintense stimulus-units upon the discriminations of *Lr* and of *F*. Hull's K, or incentive motivation, corresponds to my valence, v_f. And his ${}_sH_R$, or habit, is an *S-R* connection whereas my expectancy is conceived as an activated (S_1-R_1-S_2) expectation. Further, however, it is to be noted that Hull's equation does not contain the second part of the right-hand side of my equation. It may be recalled, however, that he does introduce, separately, the construct of conditioned inhibition ${}_sI_R$ which, if I interpret him correctly, is somewhat related to what I am saying about w or expected work. ${}_sI_R$ he apparently subtracts from ${}_sE_R$ just as I subtract $f_y\ (n_{-w},\ v_{-w},\ \exp_w)$. It must be pointed out, however, that for Hull inhibition is somewhat different from my concept of expected work and need against work. But in any event, both he and I assume that the "expected work" or the "conditioned inhibition" gets established through repetitions of the activity involved.

Turn now to escape behavior. I would feel that, in the usual case, escape is not as has been suggested by the Hullians primarily a getting-to a positive valence attached to the reduction of pain but is rather simply a getting-away from the negative valence of the shock itself. Such an interpretation is represented by Fig. 2. In Fig. 2 the jumping is represented as a mere leading away from the grid. And the animal will merely jump in any direction easily available.

Turn next to avoidance behavior. Avoidance behavior has always been something of a problem for *S-R* psychologists. This point was made some years ago by Hilgard and Marquis (8) and has recently been re-emphasized and subjected to a further analysis by

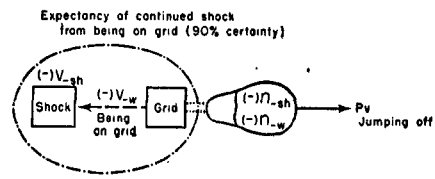


FIG. 2. Behavior-space: rat escaping from charged grid.

Ritchie (16). In avoidance behavior the performance vector results, according to my language, from the expectation that going to the grid is going to lead to shock. And this performance vector pushes or pulls the animal away from the grid even when he is not on it. The negative need-push against shock will lead, not as in the escape case merely to getting off the grid after the shock has come, but rather to the performance of avoiding the grid altogether. See Fig. 3. In such a case, pain never occurs and hence there is no pain to be reduced. Mowrer (14), of course, is the one among the stimulus-response-oriented psychologists who has most clearly seen this. And he has solved it by postulating a conditioned anxiety to the whole grid box, the reduction of which constitutes positive reinforcement. Now while I would accept something like Mowrer's interpretation for many instances, I do not believe that anxiety is always or necessarily involved. The skilled human subject who has learned to lift his finger in time to avoid a minor shock is not usually, I believe, suffused with even a minimum of anxiety. He responds so as to avoid the shock, quite unemotionally and purely on the basis of his expectancy, his need-push, and the negative valence of the shock. He expects shock in a certain place at a certain time, provided he goes to or stays in that place, and his performance is one of avoiding that place before that time, in a quite relaxed, nonanxious fashion.

Finally, however, it is to be noted that although my *diagrams* for escape and avoidance are different from that for approach, my algebraic equation for either escape or avoidance would be of the same form as that for approach (Equation 3).

$$Pv = f_w\,(n_{-sh},\ w_{-sh},\ \exp_{sh}) - f_y\,(n_{-w},\ v_{-w},\ \exp_w) \quad [3]$$

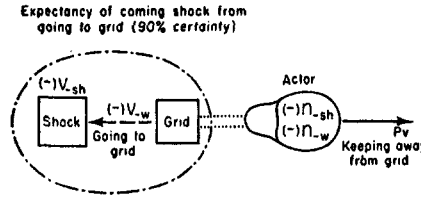


FIG. 3. Behavior-space: rat avoiding grid from which shock is expected.

The greater the need-push n_{-sh} against shock, the greater the negative valence v_{-sh} of the particular shock and the more certain the expectation $\exp_{sh}$ (or immediate discrimination of shock), the stronger will be the performance vector leading to behavior away from the sign of pain (in the case of avoidance) or leading to the leaving of the pain-object (in the case of escape). And the greater the need against work n_{-w}, the greater the negative valence v_{-w} of the particular work involved in the escape or the avoidance and the greater the expectancy of this work, the smaller will be the Pv toward escape or avoidance.

It is to be noted further, however, that, in the avoidance case, w, or the expected work involved between sign and significate, can be large and lead to relatively weak avoidance of the sign as a result of a number of relatively different physical reasons. The w may be waiting a long time for the shock. Or it may involve doing some special further act such as pressing a treadle for the shock to appear. Or, to take a quite different example, where the significate is not pain from shock but pain from a rat-hungry cat, w might involve the breaking down of a tough barrier by the rat between it and the cat. In all three cases the avoidance response of the rat will be relatively weaker than it would have been if these intervening w's had been small.

But up to now you may well feel that my formula and Hull's are not significantly different. And, if you do, so

much the better. It would indicate that there has been really much common agreement between us in spite of differences of words, phrases, and slogans.

However, when it comes to cases where not one but two or more alternative performance tendencies are involved and result in some quite new performance—such, for example, as taking a compromise path between two original ones, taking a shortcut, taking a roundabout route, VTEing, or the like—then depicting the situation by means of the behavior space becomes (to me) more helpful and suggestive. (And, of course, not to renege on my basic position, my own "private style of thinking," to quote Kendler [11], is such that I cannot easily think about the interaction of a single sign, expectancy, need-push, and valence without first seeing these in some sort of spatial terms.) But turn now to the new problem of combining two expectancies. I shall introduce here a new concept, that of the angular dispersion between expectancies.

This concept of angles is tricky and I know I am not wholly clear on it. It is one that Lewin also bogged down on. Nevertheless let me present, as an illustration, one type of compound performance which I would explain with the help of angles, namely, VTEing (vicarious trial and error). The explanation will be closely related to, but in some ways different from, that which Clark Crannell and I evolved some years ago (21, 22) in connection with what I then called the "schematic sowbug." Actually, anyone who read the earlier papers may have already recognized that the actor as I have presented him here is no more than a modern, and I hope a less complicated, version of my old friend, the schematic sowbug.

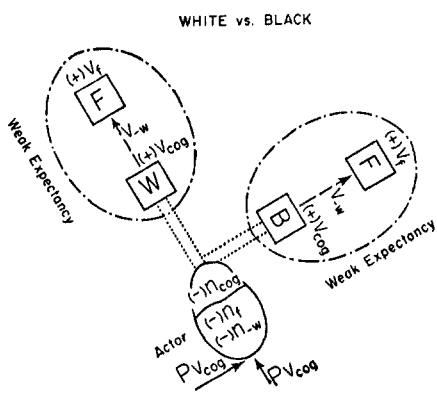


FIG. 4a. Behavior-space, early in discrimination learning. White card versus black card. Much VTE.

There are two sets of empirical findings as to VTEing to be emphasized: (*a*) Early in learning, just before (or just as) the rats begin to react to the crucial sign dimension, they tend to do more VTEing when there is a large discriminable difference between the two cards than when there is a small one. They VTE more for a white card vs. a black card than for a white card vs. a light gray card (21). (*b*) Later, however, when they have had much preceding experience in discriminating along the white-gray-black dimension, they tend to VTE more when the discriminable difference between the two cards is small (24). They VTE more for white vs. light gray cards than for white vs. black cards.

Consider first the behavior-space pictures early in learning when the rats VTE more for the white-black than for the white-gray. (See Fig. 4a and Fig. 4b.)

In Fig. 4a the actor (the sowbug) is in the white-black situation and in Fig. 4b in the white-gray one. And I have assumed that white vs. black corresponds to a 90° angle relative to the actor's (sowbug's) nose, whereas the white vs. gray corresponds to a smaller angle. I have also assumed that in this early part of learning there operates, in addition to the hunger need-push n_f, a more purely cognitive sort of need-push

to examine new stimuli, n_{cog}. The need-push corresponding to this cognitive-examination drive I have indicated by a minus sign in the fore part of the actor or bug. I have also assumed that both the stimulus cards have for a while, as new stimuli, positive valences for this cognitive-examining drive. That is, there will be performance vectors pushing the actor to "look at" each stimulus card. These vectors are those indicated at the rear of the bug. In addition, I have indicated, by putting in F (for food) with a plus valence on it on both sides, that at this early stage of learning there have developed expectancies that going toward either card may lead to food. And for the diagram to be complete there should also be another pair of shortish vectors pushing the bug, as a result of these expectancies, to jump towards each card. I have omitted these latter, jumping vectors, so as not to complicate the picture. The vectors shown are "looking towards and examining" vectors.

I shall assume further (and here I am assuming very much what Glanzer [3, 4] has proposed in two recent articles in which he was concerned with alternation behavior in a symmetrical maze) that in the course of looking at one card the valence of this card as "something to look at and examine" tends to become temporarily discharged or satiated. This then weakens, for the moment, the vector for looking at this first card. The actor then turns to look at the other card. The vector toward this latter card, however, then becomes satiated in its turn; whereas the vector for looking at the first card becomes stronger again. So the rat looks back and forth, or VTE's, until satiation for both cards, as far as looking or examining is concerned, has become complete. In short, the rat VTE's until jumping takes over.

To explain further why there is more VTEing for white vs. black (Fig. 4a) than for white vs. gray (Fig. 4b) I shall also assume that since in the case of white vs. black the two cognitive-examination vectors are at right angles, the reduction of one of these vectors through satiation has no effect on the other. The rat gets temporarily satiated for white but he still wants to look at black, and vice versa. In the case of the white vs. gray, on the other hand, the two examining vectors are at a narrower, acute, angle which means that the shortening of one vector through satiation projects upon and tends to shorten the other. Looking at white reduces temporarily not only the vector to look at white but also the one to look at gray, and vice versa. Hence early in learning or just before learning there will be less VTEing for white vs. gray than for white vs. black.

Consider now the situation later in learning. The original cognitive-examination drive is assumed now to have become wholly satiated. The vectors corresponding to it will have disappeared. But expectations have begun to build

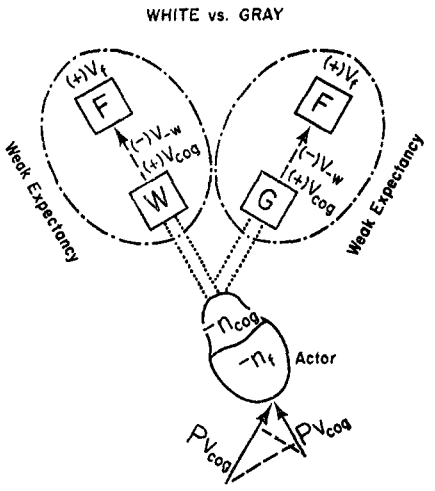


FIG. 4b. Behavior-space, early in discrimination learning. White card versus gray card. Little VTE.

up to the effect that, in the one case, white leads to food and black to no food, and, in the other, that white leads to food and gray to no food. These situations are represented in Figs. 5a and 5b.

The vectors are no longer ones for looking, but ones for jumping to white (as sign of food) and for not-jumping to black (as sign of nonfood). In the white-black case (Fig. 5a), because the two vectors are at right angles the negative vector "away from the black" does not affect the positive one "toward the white," and vice versa. In the white-gray case (Fig. 5b), on the contrary, since the two arrows are at an acute angle, the negative vector "away from the gray" does tend to project upon and reduce the positive vector "toward the white"; and the positive one "toward the white" projects upon and decreases the negative one "away from the gray." And this means in the white-gray case that until the expectancies become very certain and the vectors very long, the reduction of the positive vector by the negative one and vice versa will be sufficient to keep the rat in unstable equilibrium so that VTEing will tend to continue much longer.

Now, for the sake of the nonvisually minded critic, let me here suggest that perhaps a more respectable and conventional way of talking would be to say merely that early in learning there is more generalization of satiation of the cognitive examining drive from white to gray than from white to black; whereas, later in learning, there is more generalization of acquired approach and avoidance tendencies from white to gray than from white to black. More generalization of satiation for "looking at" early in learning means *less* VTEing between white and gray, whereas more generalization of approach and avoidance tendencies later in learning means *more* VTEing between white and gray. Actually, however, I suspect that such a more "respectable" way of talking is really just as confused and as full of hidden assumptions as is my pictorial way. In any event, the pictorial terms suggest to me (and I emphasize *to me*) a greater number of further experiments to try out than do the purely verbal ones.

The topic of this paper was "Principles of Performance." And by a per-

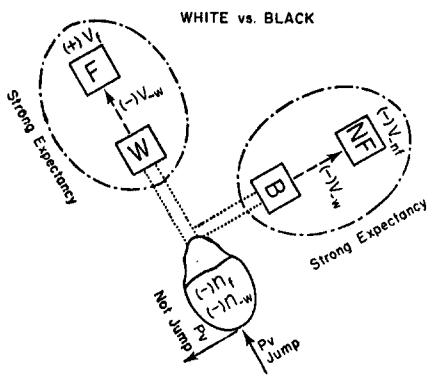


FIG. 5a. Behavior-space, later in discrimination learning. White card versus black card. Little VTE.

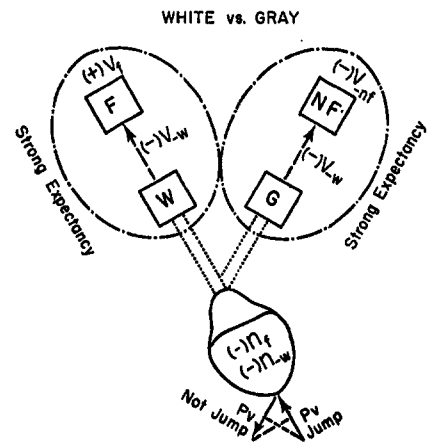


FIG. 5b. Behavior-space, later in discrimination learning. White card versus gray card. Much VTE.

formance was meant, as has been said, a generalized way of behaving to be discovered and specified only by observing the responses in at least two and perhaps more concrete test situations. Further, a performance when thus discovered is identifiable only in general terms, such as approach, consummation, escape, avoidance, vicarious trial-and-error examination, or the like. The motor skill characteristics involved in such a performance are not included in its definition. Only in a very general way may a performance be further specified as a "looking at," a "jumping to," a "running down," a "pressing," an "eating," or a "drinking." The point of view being argued for is that performances and not movements or skills are all that we ought now, at the present time, to try to explain or to predict in detail. Under certain controlled conditions we can use movements as pointer readings for performances but we cannot as yet predict movements independently of the performances within which they occur. As Brunswik has pointed out (1), all kinds of different movements and skills may vicariously substitute for one another and produce one and the same achievement or performance. Movements and skills in his terms are mere mediational, peripheral phenomena. It is the "distal," not the "proximal," characters of *S*'s and *R*'s to which I am suggesting that we devote ourselves for the time being. This is my interest and my bias.

One last word. Most of the really interesting problems of psychology—such as the distortion effects of motivation on perception (and I still believe there are some) or again those facts which have to do with what I have called the widening or narrowing of cognitive maps (23)—will eventually find their place, I believe, under the heading of complications in the principles of performance. Or, to use the language of this paper, they will turn out to be describable in terms of interesting complications and interactions between the performance vectors, as these complications and interactions appear in particular behavior-space setups.

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